

Akul Chordia

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EDUCATION

B.Tech Computer Science and Engineering	Vellore Institute of Technology, Vellore	8.31/10 (7 th semester)	2022 – 2026
Diploma, Data Science and Programming	Indian Institute of Technology, Madras		2022 – 2024
High School Diploma	Rockwoods High School, Udaipur	91.4%	2017 – 2022

WORK EXPERIENCE

Raise Financial Services - Software Engineer Intern

- Built and deployed a high-throughput API using SSE, handling 10k+ req/s, with cached snapshots for low-latency reads and MongoDB for long-term historical storage.
 - Developed an internal compliance service to automatically tag associated creators based on content, supporting policy enforcement workflows.
- Jan 2026 – Current

EaseLearn AI - Head of AI

- Cut query latency 40% by implementing a scalable routing pipeline with asynchronous processing and optimized model selection logic; system has 25k+ users.
 - Reduced jailbreak vulnerabilities by ~85% by engineering a multi-stage defense pipeline that utilized prompt analysis, fine-tuned guardrail models, and topic restriction to create an enterprise-grade secure AI infrastructure.
- July 2025 – Dec 2025 [link](#)

NSE Indices Ltd. - Quantitative Research Intern

- Collaborated on algorithmic index design, development and back testing of prototype indices, evaluating performance and market fit;
 - Benchmarked multiple providers to identify white-space opportunities in index design; directly contributed to strategy for new product lines where NSE had no presence.
 - Delivered final presentation on findings and recommendations to the CEO, VPs, and Director of NSE Indices, shaping senior leadership's roadmap for future index offerings.
- May 2025 - July 2025 [link](#)

Particle Black - Data Analyst Intern

- Engineered a machine learning pipeline using scikit-learn to predict real-estate sales, encompassing data ingestion, model training, and validation; application in property valuation and credit risk modeling, data analysis;
 - Built features from raw listing data (location, price, attributes, historical sales); framework extendable to large-scale asset pricing.
- Feb 2025 – May 2025 [link](#)

PROJECTS

OrderBook (C++)

- Built a high-performance agent based multi-threaded order book with OOPs based modular architecture
 - Simulated realistic exchange microstructure by enabling various concurrent agents including zero-intelligence agents, market makers, TWAP, VWAP agents and HFT agents with dynamic quoting and spread, inventory skew, and general risk management.
 - Currently working on further academic research with a professor.
- [github](#)
[whitepaper](#)

CPU Simulator (C++)

- Engineered a functional CPU from first principles using only AND/NOT gates; implemented custom assembly language.
 - Implemented a comprehensive CLI with program loading, step-by-step debugging, register status monitoring, and execution control
 - Foundation for educational tools, hardware simulation, or low-level system research.
- [link](#)

Exon Detection (Python)

- Built several types of CNN optimization algorithms (PSO, PO, RIME, AVOA) for the detection and classification of Exon in human DNA
 - Currently in the process of getting a research paper approved
- [whitepaper](#)

Narrative Game Engine (C++)

- Ported Windows-based OpenGL engine to macOS reworking windowing, input, and rendering backends.
 - Project illustrates adaptability in systems-level engineering; foundation for future cross-platform engines
- [link](#)

Others

[Options Pricing](#) - Black-Scholes, Binomial models and Monte Carlo Simulations (C++) | [option payoff graph](#) (Python)
[Auction-style price discovery simulation](#) - (Python) | [Statistical Arbitrage](#) - Pairs Trading Strategy (Python)
[Shopeasy](#) - Real-time cart retail system (Flutter) | [Betr](#) - custom contract betting (Django) | [FOCUS](#) - LLM based ADHD diagnosis (Flask)

TECHNICAL SKILLS

Programming Languages - C++ (Advanced), Python (Advanced), Ruby, C, SQL, JavaScript, Java, Go

Computing and Software Engineering - Data Structures & Algorithms, API Integration, Low-Latency Data Pipelines, Model Development

Systems & Tools - Linux, Unix, Git, GitHub, Validation Testing, Model Documentation, Shell Scripting, CI/CD, HTTP requests

Finance, Data and Analytics - NumPy, SciPy, Quantitative Reasoning, AWS, Database Management Systems, Linear Algebra, Statistical Modeling

Development and Productivity - Testing & Debugging, Excel, PowerPoint, Automation Scripts, Performance Optimization, Oral communication

EXTRA CURRICULARS

4x Hackathon winner (VIT Central Hack, Electra 2025, IIT Madras DeepRacer, ACM C/C++ challenge, etc)

VIT Debate Club - Senior Core - Led one of India's largest debate clubs; organized and competed in global tournaments; Mentored and judged school teams; developed a scalable outreach model for competitive debate training.